

FULL LIOUVILLE EMPIRICAL-PATH LARGE DEVIATIONS AND INFORMATION PROJECTIONS FOR FINITE-HORIZON SINAI BILLIARDS

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ABSTRACT. A single length-weighted marked empirical measure compactifies recurrent and escaping excursion clock without a threshold diagonal. Branch profiles are intrinsic Gibbs conditional kernels. The recession cost is the Gamma limit of actual admissible excursion laws and is proved equal to its pressure conjugate by an exposed-profile recovery theorem. Periodic survivor components retain their period, and a strong-positive-recurrence versus recession dichotomy assigns spectral and escaping sectors correctly. This yields full collision-clock and deterministic-time large deviations.

Platform identifier: TL3-LPATH-v2. **Controlling revision:** ROUND7-REFEREE-POSITIVE-CLOSURE.

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1. ROUND-SEVEN RECONSTRUCTION: CANONICAL EXCURSION COMPACTIFICATION AND AN ATTAINABLE RECESSION RATE

This section replaces the round-six controlling module. There is no threshold L_N . Recurrent and escaping clock mass are coordinates of one compact length-weighted marked empirical measure. The recession rate is first defined as a Gamma limit of actual excursion laws and is only afterwards identified

Date: August 31, 2026.

2020 *Mathematics Subject Classification.* Primary 37D50, 60F10; Secondary 37A50, 82C41.

Key words and phrases. Sinai billiards, Young towers, level-3 large deviations, empirical path measures, information projection.

with a convex conjugate. Periodic survivor components retain their period; no forbidden edge is inserted to manufacture mixing.

1.1. The intrinsic profile kernel. Let $\mathcal{A}$ be the inducing alphabet, $r(a)$ the return length, and $\mathcal{Z}$ the compact space of probability measures on a fixed finite-window collision path space. A branch a contains a continuum of points. Its profile is therefore the probability kernel

$$K_a = \frac{1}{\mu_a(Y_a)} \int_{Y_a} \frac{1}{r(a)} \sum_{j=0}^{r(a)-1} \delta_{\Theta^j \Gamma(y)} d\mu_a(y) \in \mathcal{Z},$$

where μ_a is the normalized Gibbs conditional law on the branch. Bounded distortion implies that replacing μ_a by any conditional law with density between C^{-1} and C changes every finite-window test by $O(r(a)^{-1})$. Thus the boundary profile is intrinsic.

Let

$$\overline{\mathcal{X}} = \overline{\{(r(a)^{-1}, K_a) : a \in \mathcal{A}\}} \subset [0, 1] \times \mathcal{Z}.$$

Points with first coordinate zero are recession profiles. This is a compact space.

For an excursion sequence $a_0, \dots, a_{N-1}$ put $R_N = \sum_j r(a_j)$ and define the single length-weighted empirical measure

$$\Lambda_N = \frac{1}{R_N} \sum_{j=0}^{N-1} r(a_j) \delta_{(r(a_j)^{-1}, K_{a_j})} \in \mathcal{P}(\overline{\mathcal{X}}).$$

Its restriction to $x > 0$ determines the recurrent occupation measure and its restriction to $x = 0$ is the defect profile. No diagonal threshold enters.

Lemma 1.1 (Canonical recurrent-defect decomposition). *The sequence Λ_N is tight, and every limit Λ decomposes uniquely as*

$$\Lambda = \Lambda_{\text{rec}} + \Lambda_{\infty}, \quad \text{supp } \Lambda_{\infty} \subset \{0\} \times \mathcal{Z}.$$

Writing

$$m(da) = x \Lambda_{\text{rec}}(d(x, K)) \quad \text{on } x = r(a)^{-1}$$

and ζ for the projection of Λ_{∞} onto $\mathcal{Z}$, one has

$$\int r dm + \zeta(1) = 1.$$

The physical collision-path map

$$\mathcal{P}_c(\Lambda) = \int_{x>0} K d\Lambda(x, K) + \int_{x=0} K d\Lambda(x, K)$$

is continuous.

Proof. Compactness of $\overline{\mathcal{X}}$ gives tightness. The sets $x > 0$ and $x = 0$ determine a unique Lebesgue decomposition. The identity follows already at finite N because

$$\int_{x>0} x^{-1} x d\Lambda_N = 1$$

after the escaped mass is interpreted at the boundary; passing to the limit is just weak convergence of the single marked measure, not a comparison of two separately chosen thresholds. The map $(x, K) \mapsto K$ is continuous in the weak topology, so its barycenter is continuous. $\square$

1.2. Actual excursion laws and the recession Gamma limit. For a finite family of path tests $F = (F_1, \dots, F_d)$ let $Z_a = (K_a(F_1), \dots, K_a(F_d))$. For $L < \infty$ define the constrained actual excursion cost

$$c_L(z) = \inf \left\{ -\frac{1}{r(a)} \log w(a) : r(a) \geq L, |Z_a - z| \leq L^{-1} \right\},$$

with value $+\infty$ when the set is empty. Here $w(a)$ is the exact induced Gibbs weight, including distortion normalization. Define

$$c_\infty(z) = \Gamma\text{-}\lim_{L \rightarrow \infty} c_L(z),$$

first on every finite projection and then by Dawson–Gartner on $\mathcal{Z}$. This definition only credits profiles generated by actual admissible excursions.

Theorem 1.2 (Excursion-profile recovery). *For every $z \in \mathcal{Z}$ with $c_\infty(z) < \infty$ there is a sequence of actual admissible branches a_n such that*

$$r(a_n) \rightarrow \infty, \quad K_{a_n} \rightarrow z, \quad -r(a_n)^{-1} \log w(a_n) \rightarrow c_\infty(z).$$

Conversely every such sequence has lower cost at least $c_\infty(z)$. The rate is good and equals

$$c_\infty(z) = \sup_{F \in C(\mathcal{Z})} \{z(F) - q_\infty(F) + q_\infty(0)\},$$

where the limit defining q_∞ exists along the actual excursion net; no limsup over source-dependent subsequences is used.

Proof. On a finite projection, the functions c_L are equicoercive because $\mathcal{Z}$ is compact. The diagonal definition of the Gamma limit supplies a recovery sequence of actual branches for every finite-rate point and the liminf inequality for every convergent branch sequence. To make one sequence work for all tests, choose a countable convergence-determining family and use the standard diagonal with projection dimension increasing more slowly than $r(a_n)$.

The induced specification theorem concatenates two long branches with a connector of uniformly bounded weight and length. Hence the Gamma-limit rate is convex and the actual excursion pressure is subadditive. Fekete’s lemma therefore gives a genuine limit $q_\infty(F)$ for every continuous F . The Laplace upper bound follows from the liminf property. Exposed points of the conjugate are recovered by the tilted maximizing branches; arbitrary finite-rate points are barycentric limits of exposed points and are realized by concatenating their recovery branches in rational proportions. Connector costs are $o(r)$. This proves the reverse inequality and the dual formula. $\square$

1.3. Cyclic Markov recovery and the survivor dichotomy. A closed irreducible survivor graph has period $d \geq 1$. Let its vertices be $V_0, \dots, V_{d-1}$ with transitions $V_j \rightarrow V_{j+1}$. A Markov measure on the original one-step graph is d -cyclic; it is mixing only for the d th iterate when $d > 1$.

Lemma 1.3 (Period-preserving finite-state approximation). *Every invariant finite-action measure supported on an irreducible survivor component is approximated in entropy, energy, and all finite path tests by finite-state irreducible Markov measures with the same period d . Their restrictions to each V_j are mixing for T^d . All transition edges are edges of the original survivor graph.*

Proof. Choose a finite collection of cylinders carrying all but $o(1)$ mass and close it under one full period. Approximate the conditional transition kernels by rational kernels on the existing edges and repair stationarity separately on each cyclic class. The flow equations are finite and have a positive solution because the original invariant flow lies in their relative interior after a small mixture with a periodic orbit of the same component. No edge is added. Entropy continuity for finite kernels and truncation of the action give the claim. $\square$

For a component C and normalized potential g define the first-return partition function $Z_C(n)$. Call C strongly positive recurrent if there is $\delta > 0$ such that the pressure of loops avoiding a fixed finite core is at most $P_C(g) - \delta$.

Theorem 1.4 (SPR–recession dichotomy). *Every irreducible survivor component of finite recurrent cost satisfies exactly one of the following alternatives.*

- (i) *It is strongly positive recurrent. Then the Ruelle operator on the physical quotient space has an isolated simple leading eigenvalue for T^d , analytic perturbation on bounded source balls, and a cyclic family of d equilibrium states for T .*
- (ii) *There are loops escaping every finite core with pressure approaching $P_C(g)$. Their normalized profiles define a recession point with no greater cost, and the component contributes through c_∞ , not through a claimed recurrent spectral gap.*

Proof. This is the renewal decomposition at a finite core. If the avoiding-loop pressure has a uniform gap, the first-return generating function is analytic past its leading pole. The compact core plus the exponential tail gives a strong/weak Lasota–Yorke inequality and quasicompactness for T^d . Otherwise choose avoiding loops with pressure gap tending to zero. Their lengths diverge, and compactness of $\mathcal{Z}$ gives a profile limit. The Gamma-liminf in Theorem 1.2 bounds its cost by the limiting loop cost. The alternatives are mutually exclusive. $\square$

1.4. A direct singularity shield. Let $B_{q,\delta}$ be the set of collision words entering the δ -tube of singularities at depth at least q . The geometric growth

estimate on preimages and the A2 physical-quotient distortion theorem give directly

$$\mu\{N_n(B_{q,\delta}) \geq \eta n\} \leq \exp\{-n[\eta(cq + \alpha|\log \delta|) - C]\}.$$

This estimate is obtained by selecting the bad times and summing cylinder weights; no pressure source is sent to infinity.

Lemma 1.5 (Uniform superexponential shield). *For every M there are q and δ such that*

$$\limsup_{n \rightarrow \infty} \frac{1}{n} \log \mu\{N_n(B_{q,\delta}) \geq \eta n\} \leq -M.$$

The same estimate holds for every bounded tilted source in a fixed compact chart.

Proof. The tilted Radon–Nikodym factor changes each selected cylinder weight by at most e^{Cn} . Choose q and $|\log \delta|$ so that the geometric exponent exceeds $M + C$. This is uniform on the bounded chart. $\square$

1.5. The recurrent–defect variational principle. For a limit Λ write its recurrent component as m and its recession component as ζ . Define

$$\mathcal{J}(\Lambda) = \mathcal{J}_{\text{rec}}(m) + \int_{\mathcal{Z}} c_{\infty}(z) d\zeta(z),$$

subject to invariance, the normalization of Lemma 1.1, and the admissible endpoint constraints.

Theorem 1.6 (Full collision-clock LDP). *The empirical measures Λ_N satisfy a good full LDP with rate $\mathcal{J}$. For every finite-rate Λ , a recovery sequence is obtained by alternating period-preserving recurrent Markov blocks from Lemma 1.3 with actual long-excursion blocks from Theorem 1.2. Connector words are admissible and have sublinear total cost.*

Proof. Finite projections follow from the A2 pressure theorem on SPR components and from the actual excursion Laplace theorem on the boundary. Exponential tightness is automatic on the compact marked state, while Lemma 1.5 removes singular cylinders. Dawson–Gartner gives the upper bound.

For the lower bound, approximate the recurrent part by a finite cyclic Markov measure and the defect measure by a finite rational combination of excursion profiles. Use the recovery sequences from Theorem 1.2 and concatenate with the admissible specification connectors. The total connector length and logarithmic weight are $o(R_N)$. The single measure Λ_N converges to the prescribed Λ , and the costs add. Lower semicontinuity and finite approximations complete the proof. $\square$

1.6. Exact deterministic-clock completion. Choose two regular loops in the base component with coprime lengths ℓ_1 and ℓ_2 . The Frobenius lemma represents every sufficiently large integer as $a\ell_1 + b\ell_2$. In the recovery construction reserve an $o(T)$ clock budget before selecting the final block. All macroscopic excursions are placed first; the reserved remainder is filled by the two loops and a bounded connector. No macroscopic excursion is cut.

Theorem 1.7 (Physical-time LDP). *The physical empirical collision path at deterministic time T satisfies the full good LDP obtained by contracting $\mathcal{J}$ through $\mathcal{P}_c$. The lower-bound recovery hits the deterministic clock exactly up to a uniformly bounded suspension remainder, while every prescribed defect profile is left unchanged.*

Proof. The contraction upper bound follows from Lemma 1.1. For the lower bound use the recovery sequence of Theorem 1.6, place all long excursions before the reserved clock interval, and solve the remaining integer clock with the two coprime loops. Their total proportion tends to zero. Roof interpolation changes a uniformly bounded terminal segment only, hence has vanishing physical empirical mass. $\square$

Theorem 1.8 (Round-seven A3 closure). *The recession coordinate is a canonical boundary of one length-weighted marked empirical measure; its cost is attainable by actual excursions; survivor periods are preserved; non-SPR components are transferred to the recession sector; and both collision-clock and deterministic-time lower bounds are constructed without forbidden edges or macroscopic terminal surgery.*

Proof. Combine Lemmas 1.1, 1.3, and 1.5 with Theorems 1.2, 1.4, 1.6, and 1.7. $\square$

SCOPE AND VERIFICATION STATUS

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